

Ayush R David

B.Tech. Computer Science & Engineering –
Artificial Intelligence

Karunya Institute of Technology and Sciences, Coimbatore

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SUMMARY

Final-year B.Tech. AI student and research intern at ARTPARK, IISc, specialising in computer vision, deep learning, and edge AI. Proven track record in satellite image super resolution, real time CCTV analytics, and autonomous systems with national hackathon placements (ISRO BAH 4th, SIH Finalist). NVIDIA certified across 7 deep learning and GPU computing courses. Seeking opportunities in AI perception, computer vision research, and intelligent systems.

EDUCATION

Degree	Institute / Board	CGPA / %	Year
B.Tech. CSE – AI	Karunya Institute of Technology and Sciences, Coimbatore	8.1	2023–2027
Class XII (ISC)	Sherwood College, Nainital	86%	2023
Class X (ICSE)	Sherwood College, Nainital	87%	2021

ARTPARK @ IISc (AI & Robotics Technology Park)

Research Intern – Computer Vision & Intelligent Systems

Bengaluru, India
Jan 2026 – Present

- Built a real-time video analytics platform with a Kafka-based streaming pipeline, multi-model inference stack (YOLO, Insight-Face, VLMs) optimised via ONNX Runtime and vLLM, and scene-change filtering to reduce inference load on edge hardware.
- Developed an AI Copilot with local-first LLM orchestration (Ollama + Llama 3.2-3B), MCP-inspired query routing, and a RAG pipeline over Qdrant – fully air-gapped with no external API dependency.
- Engineered a production full-stack dashboard (FastAPI + React) with WebSocket live feeds, MJPEG/WebRTC stream rendering, and CI/CD via GitHub Actions with Docker Compose orchestration across all platform services.

Karunya Innovation and Design Studio (KIDS)

Computer Vision Engineer Trainee

Coimbatore, India
2024 – Present

- Developed and deployed 5+ real-time computer vision systems on edge hardware (Jetson Nano, ZED2i); led occupancy analytics pipeline serving 10+ live CCTV feeds across campus infrastructure.
- Designed and delivered Arduino training bootcamp as instructor; mentored students in embedded AI and CV fundamentals.

NVIDIA Student Network – Karunya

Chapter Lead

Coimbatore, India
2024 – Present

- Established the NVIDIA Deep Learning Institute (DLI) program presence on campus; trained 30+ students in GPU computing, deep learning, and edge AI deployments on Jetson hardware.

Institution Innovation Council (IIC) – Karunya

Student Convener (2025) | Student Coordinator (2024)

Coimbatore, India
2024 – Present

- Coordinated innovation initiatives, inter collegiate events, and mentored junior students on ideation and prototype development.

PROJECTS

- Optical-Guided Thermal Super-Resolution | SIH 2025 Finalist (ISRO Track)**

Physics-constrained, self-supervised $2\times-4\times$ thermal SR framework with causal optical guidance via gated FiLM fusion

Jan 2025 – Present

 - Architecture:** Multi-encoder ResUNet with thermal, structural (PAN), and physics branches; differentiable Planck law + PSF radiative forward model enforces LR thermal consistency without requiring HR ground truth.
 - Training:** Self supervised via Wald-style synthetic degradation on large scale Landsat 8 optical thermal data; hybrid loss combining radiance domain consistency, L1 temperature error (Kelvin), and gated SSIM structural loss.
 - Results:** Stable $2\times-4\times$ upscaling; evaluated on RMSE (K), PSNR, SSIM; identified and mitigated emissivity-bias and optical-hallucination failure modes.
 - Stack:** PyTorch, CUDA, OpenCV, RasterIO, Physics-Informed ML, Landsat 8 optical thermal data fusion.
- Dual Image Super Resolution for Satellite Imagery | ISRO BAH 2025 – 4th Place Nationally**

Fused two LR captures into a $256\times256 \rightarrow 512\times512$ HR image with integrated Blind Image Quality Assessment (BIQA)

Jul 2025

 - SR System:** ShiftNet sub-pixel registration + recursive HRNet fusion; achieved **41.45 dB PSNR, 0.97 SSIM** on official ISRO test set; custom 16-bit TIFF DataLoader with anti-aliasing upsampling.
 - BIQA System:** ViT + ResNet50 hybrid trained on 17,000+ image pairs; Optuna hyperparameter optimisation (HPO); training protocol: AdamW, ReduceLROnPlateau, gradient clipping, early stopping, mixed-precision.
- AI Driven CCTV Automation for Security & Forensic Analysis**

Semantic-first pipeline for real time video forensics and structured natural language query retrieval at scale

Jul 2025

 - Engineered a “semantic-first” pipeline where a PLM (Llama) generates contextual guidance for a Vision Transformer (ViT), improving both computational efficiency and analytical accuracy.
 - Designed “Semantic Feature Records” enabling structured forensic queries in natural language over large video archives; distributed microservices via Kubernetes + Kafka for high-volume RTSP stream processing.
 - Stack:** ViT, Llama, Kubernetes, Apache Kafka, Ts Db, PyTorch, OpenCV.
- NVIDIA Occupancy Analytics**

Real-time campus occupancy tracking using live CCTV; production deployment across university infrastructure

Feb 2025

 - Ingested RTSP streams via NVIDIA DeepStream; Kafka $\rightarrow$ PostgreSQL data pipeline; Grafana dashboard for live analysis; containerized deployment with Docker across 10+ camera feeds.

- **HIL Vehicle Control Simulation Bench | CARLA + Autoware AV Stack**
In-house HIL test bench for throttle/steering/braking with end to end AV integration via Zenoh middleware
 - Simulated CAN Bus vehicle control messages and actuator HIL integration; CARLA real time visualisation; validated Autoware planning stack end-to-end for autonomous driving research.

2024

- **Cross-Modal Sensor Fusion Transformer – Anti-Drone System**
Multi-sensor fusion model for robust drone detection with reduced false positive rate
 - Integrated heterogeneous sensor streams using a transformer-based fusion architecture; improved detection robustness in adversarial and cluttered environments.
- HACKATHON ACHIEVEMENTS
- **Team Lead – ISRO BAH 2025**
4th Place Nationally – Dual image super resolution & BIQA for satellite imagery
 - **Team Lead – Smart India Hackathon 2025 (SIH)**
National Finalist, ISRO Track – Optical Guided Thermal SR; physics gated fusion
 - **IISc Code Bot Hackathon 2025**
Finalist – AI auto labeling and forensic CCTV search tool
 - **Robo Racer ICRA Hackathon**
Developer – Perception, path planning & obstacle avoidance for 1:10 autonomous race cars
 - **CUMTA Hackathon**
Team Lead – AI traffic monitoring prototype using real time computer vision analytics
 - **Bashani Maha Hackathon**
Developer – Cross modal sensor fusion transformer for anti drone detection systems

2025

2025

2025

2024

2024

2024

TECHNICAL SKILLS

- **Languages:** Python, C/C++, CUDA, Java, SQL, HTML/CSS/JavaScript
 - **Deep Learning / CV:** PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, ONNX, TensorRT, TAO Toolkit, DeepStream, Physics-Informed ML
 - **Systems & DevOps:** Docker, Kubernetes, Apache Kafka, PostgreSQL, Grafana, Git, Linux, Jupyter Notebook, Google Colab

- **Autonomous Systems:** CARLA, Autoware, Isaac Sim, ROS2, F1TENTH, Zenoh middleware
 - **Hardware & Embedded:** NVIDIA Jetson Nano / Orin, ZED2i Stereo Camera , Intel Real Sense Depth Camera , CAN Bus, HIL Systems, Arduino
 - **Data & Remote Sensing:** RasterIO, Landsat 8 optical thermal data, TIFF processing , QGis , RGis ,SNAP

CERTIFICATIONS & RECOGNITION

NVIDIA DLI (7×): Fundamentals of Deep Learning | Getting Started with AI on Jetson Nano | Building Video AI Applications on Jetson Nano | Building Transformer Based NLP Applications | Fundamentals of Accelerated Computing with CUDA Python | Fundamentals of Accelerated Computing with CUDA C/C++ | Applications of AI for Predictive Maintenance

CISCO (3×): Programming Essentials in C | Programming Essentials in Python | Cybersecurity Essentials

IBM (2×): Getting Started with Enterprise Grade AI | Getting Started with Enterprise Data Science